

Hawkeye EQMS · Management Review (MRM) · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Management Review (MRM)
Pharma purpose	Periodic top-management review of QMS performance per ISO 9001:2015 §9.3 — audit results, CAPA status, deviations, complaints, supplier performance, resource adequacy, improvement opportunities, decisions, action items.
Compliance basis	ISO 9001:2015 §9.3 (management review) · 21 CFR Part 11 (e-signed minutes) · ICH Q10 §3.2.4 (KPI monitoring)
Hawkeye routes	/management-review (frontend) · /api/management-reviews (backend)
Hawkeye model file	backend/src/models/ManagementReviewModel.js
Tenant module flag	modules.MANAGEMENT_REVIEW

MRMs follow PLANNED → IN_PROGRESS → COMPLETED (or CANCELLED). The chair schedules the meeting, attendees join, the minutes-taker records inputs[] (audit / CAPA / deviation / complaint / supplier topics) and outputs (qmsAdequacy + resourceDecisions + improvementOpportunities + actionItems[]). The Wave 2 AI MRM Input Populator can auto-aggregate the last N days of cross-module data into the inputs[] array with structured summaries + trend tags + a 250-word AI narrative.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Scheduled review cadence (annual / quarterly / ad-hoc) ISO 9001 §9.3.1	reviewType enum: ANNUAL / QUARTERLY / AD_HOC / POST_INCIDENT / REGULATORY.	MET	
Inputs per §9.3.2 (audit results / CAPA / deviations / complaints / supplier perf / resource / risks) ISO 9001 §9.3.2	inputs[] subdocument array. Each: topic / summary / dataRefs[] / trend (IMPROVING/STABLE/DECLINING/NOT_ASSESSED). AI populator pulls all 9.3.2 topics automatically.	MET	
Outputs per §9.3.3 (QMS adequacy + resource decisions + improvement opportunities + actions) ISO 9001 §9.3.3	qmsAdequacy enum (ADEQUATE/NEEDS_IMPROVEMENT/INADEQUATE), resourceDecisions, improvementOpportunities[], actionItems[].	MET	
Action items with owner + dueDate + status + priority ISO 9001 §9.3.3(b)	actionItems[] subdocument with description / owner / dueDate / status (OPEN/IN_PROGRESS/COMPLETED/OVERDUE/CANCELLED) / priority / completedAt / notes.	MET	
Auto-numbered review (MRM-YYYY-NNNN) GMP traceability	reviewNumber pre-save generator.	MET	
AI auto-population of inputs from last N days of QMS data FDA AI guidance Jan 2025	POST /api/ai/mrm/populate-inputs (Wave 2). Aggregates cross-module KPIs + LLM narrative.	MET	
Minutes-taker role + e-signed minutes document 21 CFR Part 11 §11.50	minutesTakerId field; minutesDocumentId links to a published controlled document. E-sig generic endpoint exists, not yet enforced on /complete.	PARTIAL	Wire /api/electronic-signatures/sign on /complete
Chair + attendees recorded with role/title ISO 9001 §9.3.2	chairId + attendeeIds[] (User refs). Names resolved at render-time.	MET	
OVERDUE action item escalation ISO 9001 §9.3.3(b)	Status enum supports OVERDUE; no scheduler today.	GAP	Add cron that flips status when dueDate passes
Cross-module KPI rollup from /api/quality/kpis ISO 9001 §9.1	GET /api/quality/kpis aggregates audit/CAPA/deviation/complaint/training/equipment counts for arbitrary date range. Used by AI populator.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
VP Quality Elena tenant_admin · chairperson	<code>vp.quality@novex-pharma.demo</code>	Schedules + chairs MRMs, approves outputs + action items, signs minutes.	PLANNED IN_PROGRESS COMPLETED
Head of QA James admin · attendee	<code>qa.head@novex-pharma.demo</code>	Reports CAPA + deviation backlog metrics; takes action items.	IN_PROGRESS
Marcus Brown Regulatory Affairs · attendee	<code>regulatory@novex-pharma.demo</code>	Reports regulatory status + change control pipeline.	IN_PROGRESS
Audit Program Mgr Priya buyer · attendee	<code>audit.program@novex-pharma.demo</code>	Reports supplier audit programme + findings backlog.	IN_PROGRESS

4. Feature catalogue (click-by-click)

4.1 MRM register

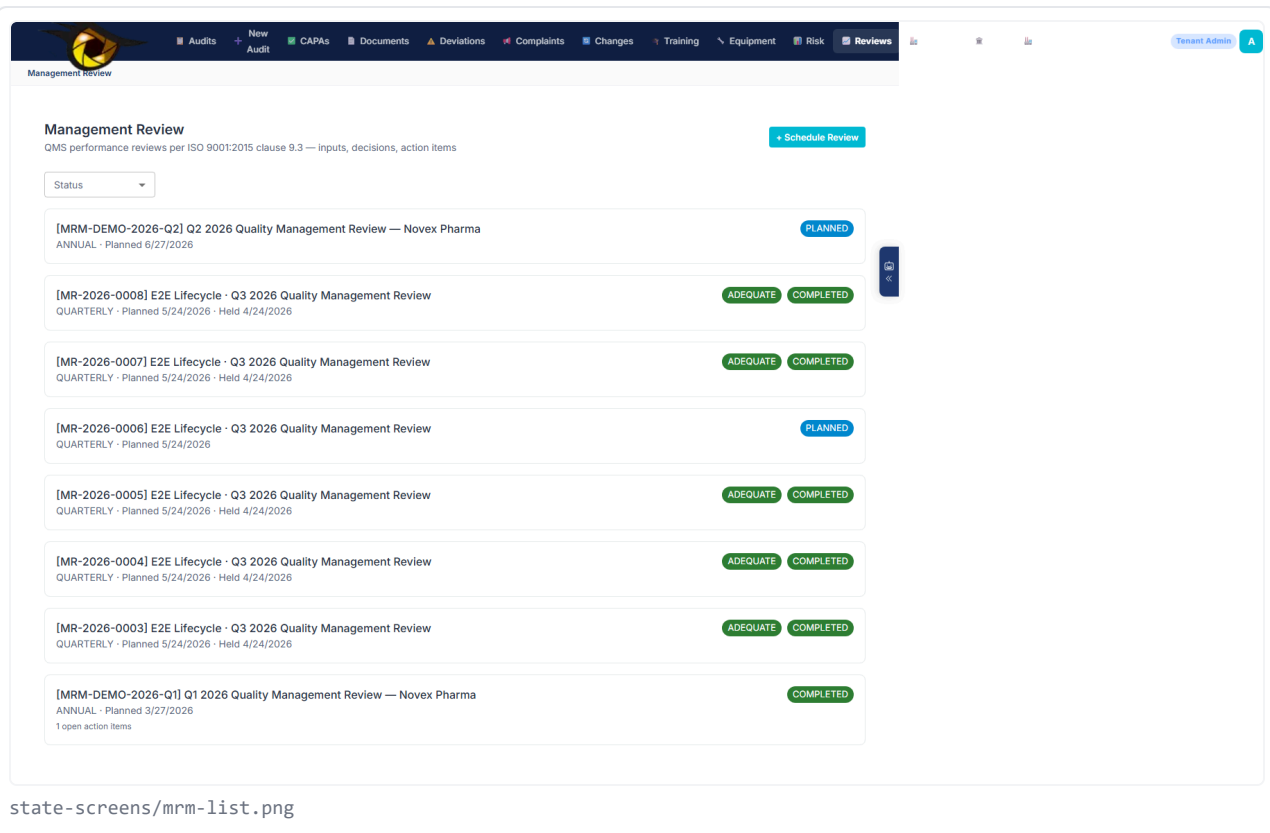
What it does	Lists scheduled + completed MRMs with reviewNumber + status chip + action-item count.
Menu / route	<code>/management-review</code>
Available to roles	any tenant viewer
Backend API	<code>GET /api/management-reviews</code>

CLICK-BY-CLICK

- NAVIGATE

 Click 'Reviews' in the top nav
→ Page renders
- WAIT

 Spinner clears
→ 2 seeded MRMs: MRM-DEMO-2026-Q2 PLANNED + MRM-DEMO-2026-Q1 COMPLETED (with '1 open action items' note)



4.2 + Schedule Review dialog

What it does	Schedule a new MRM. Captures title, type, plannedDate, chair, attendees.
Menu / route	/management-review · top-right '+ Schedule Review' button
Available to roles	chair / tenant_admin
Backend API	POST /api/management-reviews

CLICK-BY-CLICK

- CLICK** Click '+ Schedule Review'
→ Dialog opens
- TYPE** Fill title (e.g. 'Q3 2026 Quality Management Review')
→ required
- CLICK** Pick reviewType (ANNUAL / QUARTERLY / AD_HOC / POST_INCIDENT / REGULATORY)
→ default QUARTERLY
- CLICK** Pick plannedDate
→ required (must be future)
- CLICK** Pick chair from user dropdown
→ default current user
- CLICK** (Optional) Pick attendees + minutesTaker
→ Multi-select
- CLICK** Click 'Save'
→ Row appears with status PLANNED

FIELDS CAPTURED

Field	Required	Validation / values	Notes
title	Yes	string	
reviewType	Yes	ANNUAL QUARTERLY AD_HOC POST_INCIDENT REGULATORY	
plannedDate	Yes	ISO date	
chairId	Yes	ObjectId · user	
attendeeIds	No	ObjectId[]	
minutesTakerId	No	ObjectId	

4.3 Start meeting (PLANNED → IN_PROGRESS)

What it does	Mark the meeting as in progress. Captures actualDate.
Menu / route	MRM detail · 'Start Meeting' button
Available to roles	chair / tenant_admin
Backend API	PUT /api/management-reviews/:id (status=IN_PROGRESS, actualDate=now)

CLICK-BY-CLICK

1. CLICK Open MRM detail · click 'Start Meeting'
- Status flips

4.4 AI · Auto-populate inputs (Wave 2)

What it does	Aggregate last N days of audit / CAPA / deviation / complaint / supplier / resource data + AI narrative into the inputs[] array.
Menu / route	MRM detail (IN_PROGRESS) · 'Auto-populate inputs with AI' button
Available to roles	admin · tenant_admin
Backend API	POST /api/ai/mrm/populate-inputs
AI assistance	AI Free Gemini 2.5 Flash-Lite — narrative draft. Rule engine pulls KPIs from /api/quality/kpis.

CLICK-BY-CLICK

1. CLICK Click 'Auto-populate with AI'
- Drawer opens with reviewType + windowDays slider (default 90)
2. CLICK Click 'Generate'
- After ~5-8 s: inputs[] populated with one row per ISO 9001 §9.3.2 topic + a 250-word AI narrative summary

4.5 Add input section

What it does	Manually add an input topic (if not auto-populated).
Menu / route	MRM detail · '+ Add input section'
Available to roles	chair · minutes-taker
Backend API	PUT /api/management-reviews/:id (inputs[] += entry)

CLICK-BY-CLICK

1. **CLICK** Click '+ Add input section'
→ Form opens
2. **TYPE** Fill topic (e.g. 'Audit Results' / 'Customer Complaints')
→ required
3. **TYPE** Fill summary (2-4 sentences)
→ required
4. **CLICK** Pick trend (IMPROVING / STABLE / DECLINING / NOT_ASSESSED)
→ drives the trend chip on the report
5. **TYPE** (Optional) Fill dataRefs[] (links to dashboards)
→ free-text URLs
6. **CLICK** Click 'Add'
→ Input appears in the list

4.6 Add action item

What it does	Capture an action item out of the meeting. Each has owner + dueDate + priority.
Menu / route	MRM detail · 'Action Items' tab · '+ Add action item'
Available to roles	chair · minutes-taker
Backend API	PUT /api/management-reviews/:id (actionItems[] += entry)

CLICK-BY-CLICK

1. **CLICK** Click '+ Add action item'
→ Form opens
2. **TYPE** Fill description
→ required
3. **CLICK** Pick owner from user dropdown
→ ObjectId
4. **CLICK** Pick dueDate
→ required
5. **CLICK** Pick priority (LOW / MEDIUM / HIGH / CRITICAL)
→ default MEDIUM
6. **CLICK** Click 'Add'
→ Action-item appears with status=OPEN

4.7 Complete MRM (IN_PROGRESS → COMPLETED)

What it does	Final closure of the meeting. Captures qmsAdequacy + resourceDecisions + improvementOpportunities[] + actionItems[].
Menu / route	MRM detail · 'Complete + Sign' button
Available to roles	chair · tenant_admin
Backend API	POST /api/management-reviews/:id/complete

CLICK-BY-CLICK

- 1. **CLICK** Click 'Complete + Sign'
→ Drawer with all output fields
- 2. **CLICK** Pick qmsAdequacy (ADEQUATE / NEEDS_IMPROVEMENT / INADEQUATE)
→ required
- 3. **TYPE** Fill resourceDecisions
→ free-text
- 4. **TYPE** Fill improvementOpportunities[] (one per line)
- 5. **CLICK** Confirm actionItems[] are captured (added in previous step)
- 6. **CLICK** Click 'Sign + Close'
→ Status=COMPLETED · approvedBy + approvedAt set

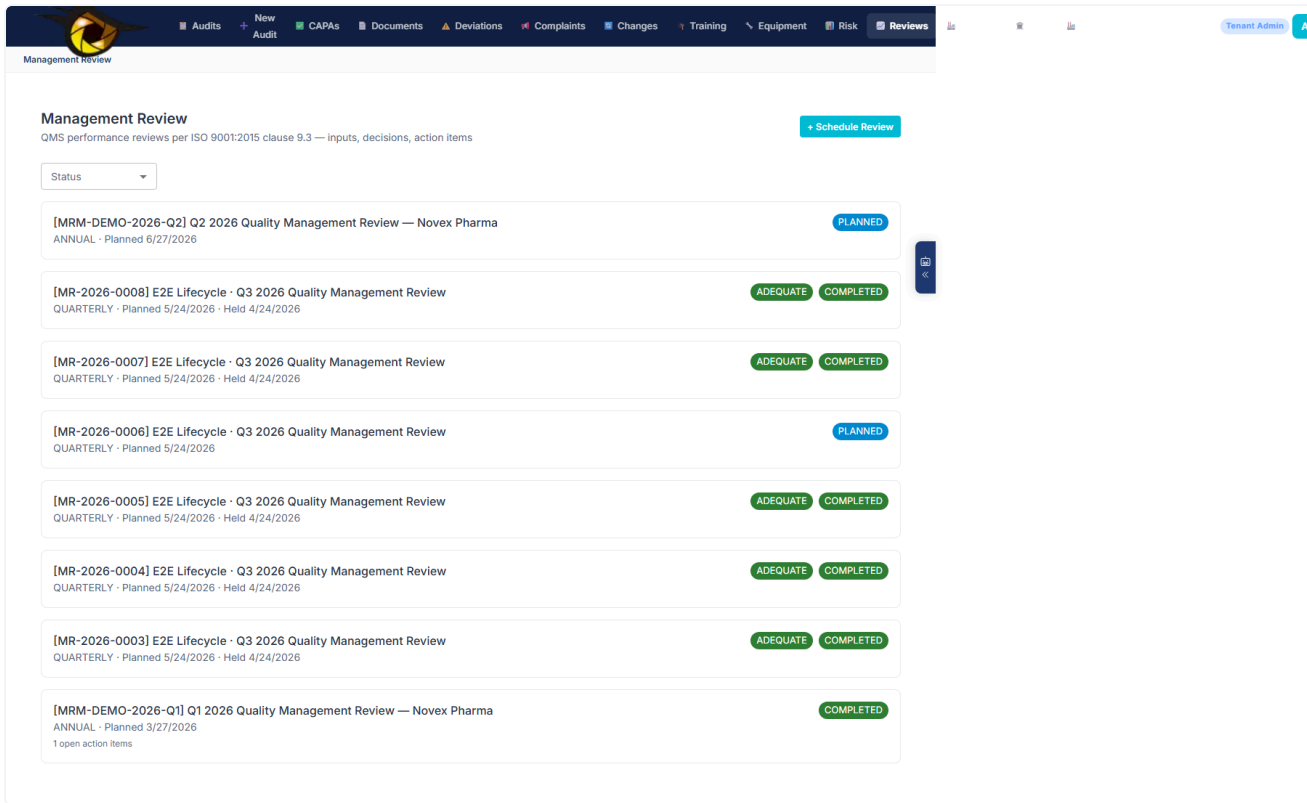
5. Full lifecycle walkthrough — one transaction, end-to-end

One MRM scheduled, started, AI-populated, completed. Persona = Elena (chair).

#	Persona	State transition	Action	API	Observed	Result
1	Elena VP Quality (chair)	— → PLANNED	Click '+ Schedule Review' → fill 'Q3 2026 QMR' / QUARTERLY / +30 days / chair=Elena → Save	POST /api/management- reviews	reviewNumber=MRM- 2026-NNNN · status=PLANNED	PASS

Expected DB state after this step:

```
management-reviews { _id, reviewNumber: 'MRM-2026-NNNN', title, reviewType: 'QUARTERLY', plannedDate: <future>, chairId: elena._id, status: 'PLANNED' }
```

#	Persona	State transition	Action	API	Observed	Result
 state-screens/mrm-list.png						
2	Elena Chair	PLANNED → IN_PROGRESS	Open MRM detail → 'Start Meeting' → actualDate set	PUT /api/management-reviews/:id (status=IN_PROGRESS, actualDate=now)	status flips	PASS
Expected DB state after this step: <pre>management-reviews { status: 'IN_PROGRESS', actualDate: <now> }</pre>						
3	Elena Chair	IN_PROGRESS → IN_PROGRESS (with AI inputs)	Click 'Auto-populate inputs with AI' → AI narrative + 6 input rows generated	POST /api/ai/mrm/populate-inputs	inputs[] populated; AI narrative ~250 words; provider=gemini-2.5-flash-lite	PASS
Expected DB state after this step: <pre>management-reviews.inputs += [{ topic: 'Audit Results', summary, trend }, { topic: 'CAPA Status', ... }, ...]</pre>						
4	Elena Chair	IN_PROGRESS → IN_PROGRESS (with action items)	Add 1 action item: 'Roll out AI 5-Why scaffolder to all QA Specialists' / priority HIGH / due +60 days	PUT /api/management-reviews/:id	actionItems[] += 1	PASS

#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step:						
management-reviews.actionItems += { description, owner, dueDate, priority: 'HIGH', status: 'OPEN' }						
5	Elena Chair + Approver	IN_PROGRESS → COMPLETED	Click 'Complete + Sign' → qmsAdequacy=ADEQUATE → resourceDecisions → improvementOpportunities[] → Submit	POST /api/management- reviews/:id/complete	status=COMPLETED · approvedBy + approvedAt set	PASS
Expected DB state after this step:						
management-reviews { status: 'COMPLETED', qmsAdequacy: 'ADEQUATE', resourceDecisions, improvementOpportunities: [...], approvedBy: elena._id, approvedAt: <now> }						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
MRM Input Populator (Wave 2)	IN_PROGRESS	POST /api/ai/mrm/populate-inputs	MRM detail · 'Auto-populate with AI'	Aggregates last N days of audit + CAPA + deviation + complaint + supplier + resource data into inputs[] with topic/summary/trend + a 250-word AI narrative	Free Gemini 2.5 Flash-Lite for narrative; rule engine for KPI aggregation
(roadmap) AI action-item suggester	IN_PROGRESS	(future)	Action Items tab	Scans the inputs[] for issues without an action item and suggests one with owner heuristic	Free Gemini

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
PLANNED	ISO 9001 §9.3.1	reviewType + plannedDate + chairId + attendeeIds[]	management-reviews
IN_PROGRESS	ISO 9001 §9.3.2	inputs[] covering all 9.3.2 topics + trend tags + dataRefs[]	management-reviews.inputs

State	Cited clauses	Evidence captured	Backing records
COMPLETED	ISO 9001 §9.3.3 21 CFR Part 11 §11.50	qmsAdequacy + resourceDecisions + improvementOpportunities[] + actionItems[] + approvedBy + approvedAt + (e-sig)	management-reviews + (electronic-signatures when wired)

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · mrm	PLANNED → IN_PROGRESS → AI populate → COMPLETED	PASS	5/5 PASS · eqms-test-results-v2.pdf
novex-walkthrough.spec.ts	/management-review register UI render with 2 seeded MRMs	PASS	18-elena.png + 20-elena.png
AI populate smoke	POST /api/ai/mrm/populate-inputs returns KPIs + narrative	PASS	Wave 2 smoke

9. Known gaps + roadmap

- **OVERDUE action-item scheduler** — Cron that flips actionItems[].status to OVERDUE when dueDate passes + sends notification. **HIGH**
- **Mandatory e-sig on /complete** — Wire /api/electronic-signatures/sign so completion writes a Part 11 signature row. **HIGH**
- **Auto-link minutesDocumentId to published doc** — When MRM completes, auto-create a controlled document of type REPORT_TEMPLATE = Minutes-of-MRM-N. Today the field exists but link is manual. **MEDIUM**
- **Visual KPI dashboard** — Bar charts for trend (IMPROVING/DECLINING) per topic. Action-item burndown. **MEDIUM**
- **AI action-item suggester** — Scans inputs[] for issues without action items. **LOW**